

Mobile Network Data To Assess Risk and Prevent Fraud

Frightening Fraud Facts

Bad actors have surged unauthorized SIM swaps by **1,055%**. Synthetic identities drain marketing budgets. And **24%** of U.S. adults fell victim to account takeover attacks in 2025.

AI powers advanced fraud that can clone real voices, generate convincing messages, and manipulate users at scale in ways that feel natural and urgent. Traditional security measures either let fraud slip or punish legitimate customers with friction that kills conversion.

Vonage Identity Insights API changes the game

Now you can raise an invisible fraud shield to plug real-time telecom intelligence — pulled from trusted carrier networks that AI cannot fake — into your risk models. Easily assess fraud risk invisibly, without adding steps to the customer journey.

Identity Insights delivers network-layer intelligence that application-layer competitors fundamentally can't access:

- Screen phone numbers
- Detect recent SIM swaps
- Validate subscriber identity against mobile operator KYC data
- Route communications intelligently

Actively stop fraud and improve deliverability

- **Account takeover (ATO):** SIM Swap Detection flags recent SIM changes in real time to block suspicious logins or trigger additional verification before attackers gain access.
- **Synthetic identity and impersonation:** Subscriber Match validates user information against mobile operator KYC data. Carrier ID flags virtual numbers and burner phones to block fake identities before they enter your system.
- **Database cleansing and communication waste:** Number Format automatically corrects and validates entries. Carrier insights identify unreachable numbers to ensure your database contains only valid, communication-ready contacts.

Get Real-Time Fraud Intelligence

Account creation

Validate phone numbers and detect suspicious patterns or mismatched data. These help block fake accounts before sign-up and reduce risk and unnecessary costs.

Login and authentication

Check for recent SIM swaps and other signals during login or password recovery to prevent account takeover attempts.

High-risk transactions

Detect elevated risk and screen customer phone data in real time to ensure legitimate users complete transactions.

Network-Powered Solutions Within Identity Insights

Enjoy plug-and-play telecom signals that enhance your risk decision engine. Use features individually or combine them to match your specific fraud prevention needs:

- **Number Format:** Cleans and standardizes phone numbers at the point of entry by fixing formatting errors before they're saved to your database. Validates against international standards and returns country codes, time zones, and format details to eliminate unnecessary costs from invalid number signups.
- **SIM Swap Detection:** Identifies when a SIM card was last changed on a mobile number with real-time checks directly from mobile operators. This allows you to block account takeover attempts before attackers intercept OTPs or access protected accounts.
- **Carrier ID:** Reduces fraud risk by verifying the current and original carrier, plus network type — such as mobile, landline, toll-free, or virtual. Avoids onboarding unreliable virtual numbers, prevents undeliverable OTPs to landlines, and ensures reachability on the most appropriate channel.
- **Subscriber Match:** Validates end-user contact information instantly — name, address, date of birth, ID document — against the mobile operator's KYC-verified data. Returns individual match scores for each attribute to mitigate synthetic identity risk and improve signup conversion rates. Costs are a fraction of formal KYC solutions.



Key Benefits

Block fraud earlier

Stop account takeovers, synthetic identities, and fraudsters before they enter your system

Reduce false positives

Leverage telco-trusted data to sharpen decision models, which protects real customers and catches bad actors.

Smooth customer friction

Add invisible risk evaluation with no extra authentication steps.

Optimize security spend

Pay only for the checks you need, when you need them — with no flat fees.

Validate before engagement

Confirm number format and carrier status before engaging. This helps eliminate wasted spend from invalid number signup attempts.

Stay compliant

Use built-in KYC-compliant identity verification through mobile operator verified records.

Start now

Learn more at:

vonage.com/communications-apis/identity-insights

Contact an expert at:

vonage.com/contact-apis